Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/23/2023
PI Name:	Bell, Suzanne Ph.D.		
Project Title:	Human Factors and Behavioral Performance Exploration Measures in HERA		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	03/01/2021	**End Date:**	01/31/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 01/31/2026 per A. Beitman/JSC (Ed., 11/13/23).		
Key Personnel Changes/Previous PI:	December 2022 report: Steven R. Anderson, PhD was added as a CoInvestigator (CoI). Per PI: Sara Whiting, Ph.D. moved from the Lab to a new role and is no longer a CoI (Ed., 4/5/23). December 2021 report: Sheena I. Dev, PhD, was added as CoI in May 2021.		
COI Name (Institution):	Landon, Lauren Ph.D. (BHP Laboratory, KBR/NASA Johnson Space Center) Dev, Sheena Ph.D. (BHP Laboratory, KBR/NASA Johnson Space Center) Anderson, Steven (BHP Laboratory, KBR/NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

NOTE (9/30/2021): Continuation of "Human Factors and Behavioral Performance Exploration Measures in HERA (previous title: Identification and Validation of BHP Standard Measures in HERA for Transport)"--PI Dr. Peter Roma, who moved to Naval Research Laboratory (NRL) in early 2021. With Dr. Roma's move to the NRL in early 2021, Dr. Suzanne Bell is now Principal Investigator.
NASA's Human Research Program (HRP) is developing a set of "Standard Measures" for use in spaceflight and analogs. These "standard measures" need to be validated in analog settings to ensure their validity, acceptability, and reliability in helping identify and measure indicators of risk associated with crew health and performance. Exploration missions are anticipated to create isolated and confined environments that will include stressors such as small teams living and working in extreme conditions for prolonged periods separated from family, friends; loss of the day/light cycle; loss or delay of communications with ground; partial gravity; and limited space, privacy, and food selection. HRP's Human Factors and Behavioral Performance (HFBP) Element seeks to maintain and enhance behavioral health and performance in such environments. The Behavioral Medicine risk (Risk of Adverse Behavioral or Behavioral Conditions and Psychiatric Disorders) is a high priority within the HRP because psychological and behavioral health factors reveal intra- and inter-individual variability. This variability poses significant challenge for measurement of risk and resilience in crews and individual astronauts during long-duration missions.

The HFBP-EM (Exploration Measures) will build upon the initial use and identification of Behavioral Core Measures (BCM), whose purpose was to evaluate the feasibility and acceptability of these outcome measures (Dinges; NNJ13ZSA002N-BMED; project title: "Standardized Behavioral Measures for Detecting Behavioral Health Risks during Exploration Missions") in high fidelity analogs. The task proposed here provides an opportunity to continue using standardized measures in order to operationalize their efficiency and effectiveness, ensure consistent measurement of these key metrics, and facilitate the analysis and understanding of these measures in assessing risk. The development and use of these expanded standardized measures are needed now to develop astronaut norms in order to determine their efficacy and ability to objectively detect, assess, and manage off nominal events; and predict future off nominal events that may compromise a mission and increase the risk of behavioral health and performance decrements. This advances our need for individualized, real-time tools that provide rapid feedback and assessment that are needed for exploration class missions. The use of BHP (Behavioral Health and Performance) Standard Measures across analogs and analog missions, allows NASA Human Research Program (HRP) to determine a baseline for assessing and monitoring specific countermeasures; and increases the construct and predictive validity of countermeasures for assessing risk and for their monitoring of behavioral health and performance in spaceflight.

Specific Aims include:

Aim 1. Provide a set of BHP standard measurements for investigators to use in proposed projects. Significance: This allows NASA HRP to streamline and make more efficient the use of multiple measurements collected on research participants during analog research and helps to ensure a reduced burden on these research participants by using a "standard" set of measures for data sharing by multiple principal investigators (PIs).

Aim 2. Enable comparison of multiple missions across spaceflight analog campaigns to quantify risk using reliable metric-based data. Significance: The "standardized measures" increase the generalization of findings across research analogs, increasing the validity and reliability of measures used to quantify, characterize, and assess the impact of spacelike analogs on prevalence of behavioral health issues, incidence rates, longer term health, and performance errors. This allows generation of reliable and sensitive metrics that can be used systematically to inform accurate risk assessment and mitigation status for future exploration spaceflight missions.

Aim 3. Provide database for data-mining and integrative modeling and increase research data quality and transfer to the Life Sciences Data Archive (LSDA). Significance: Collection of "Standardized Measures" allows for greater consistency and fidelity of data collected, enhancing the data archiving capabilities of analog outcomes, increasing accessibility of data for NASA use, metrics assessment associated with red to yellow, and yellow to green risk status, and trending, and increasing the probability that the standardized set of measures are appropriately archived in a timely manner. Although data management agreements require PIs to submit their data after completion of their research, the exact format and timeliness of that data submission varies greatly among PIs. The collection of "standardized measures" via internal directed studies helps to ensure more timely, valid, and accessible data resources to help guide risk reduction.

NOTES 2018 August: Original Principal Investigator (PI) was Thomas J. Williams, PhD, who became HRP Human Factors & Behavioral Performance (HFBP) Element Scientist shortly after the award was made; Jason S. Schneiderman, Ph.D. served as interim PI until July 10, 2017, when Peter G. Roma, Ph.D. came to Johnson Space Center and became Principal Investigator. At that time, title was changed to "Identification and Validation of BHP Standard Measures in HERA for Transport" (original title was "Identification and Validation of BHP Standard Measures").

2020 February: Project title was changed to "Human Factors and Behavioral Performance Exploration Measures in HERA" (Short Title: HFBP-EM in HERA).

Rationale for HRP Directed Research:

This research is directed because NASA must define complete scientific activities in a short time and there is insufficient time to issue a solicitation. This task is in direct response to both the August 2014 Human Subjects Review Board (HSRB) requirement and the baselined Human Research Program (HRP) Path to Risk Reduction milestone of providing standard measures to monitor crew health and performance. This will allow HRP to establish, evaluate, and manage a common set of measures for use in spaceflight and analog research to: develop baselines, systematically characterize risk likelihood and consequences, and assess effectiveness of countermeasures that work for human factors and behavioral performance risk factors. This proposal qualifies for a directed study due to the urgent, time-sensitive need to provide "standard measures" as the foundation to achieve consistent research measure for data-sharing in HERA (Human Exploration Research Analog) and to meet the highly constrained, operationally-focused data gathering and analysis that allows for greater consistency in the research methods that are specific to NASA HRP standard measures development. Additionally, the set of Behavioral Health and Performance (BHP) standardized measures in the HERA analogy reflects the more operational nature of the measures while allowing the multiple and frequent internal and external collaborations required to execute this study. The directed nature of this study also allows NASA to provide the unique research and support expertise that is needed to integrate and manage the data from all of the various participating studies to achieve HRP's intent and support to the Flight Analogs Project (FAP) within a highly constrained time schedule. Completion of the required research, support to both analog and operational requirements, and vetting of

	the evidence-based standards makes the solicitation process prohibitive. In addition, our group, the BHP Lab has already performed similar work across various analogs in support of previous NASA research and the use of the BHP Lab offers considerable efficiencies that are realized by building upon our existing work and that expertise. Finally, two panels of extramural subject matter experts with experience in critical task identification in academic, military, and spaceflight analog settings met at Johnson Space Center (JSC) and provided recommendations consistent with the enclosed project related to team processes (November 2015) and psychometric assessment of cognition (April 2016).
Research Impact/Earth Benefits:	HFBP Exploration Measures provides a comprehensive understanding of the behavioral medicine and team dynamics in isolated and confined settings. Increasingly, the American population is isolated from friends, family, and coworkers due to quarantine and remote work. In addition, there are special populations such as prisoners and nursing home residents who can encounter extended separation from family and friends. Our research provides insights into the psychological impact of isolation and confinement as well as how the circumstance can shape relational and team dynamics over time.
Task Progress:	Over the past year, the BHP Laboratory made tremendous progress in three primary areas: (1) data collection and processing from all four missions of HERA (Human Exploration Research Analog) Campaign 6; (2) continued validation of Human Factors Behavioral Performance – Exploration Measures (HFBP-EM) short form measures in HERA Campaign 6; and (3) protocol development for HERA Campaign 7. First, we made significant progress in the data collection, cleaning, processing, and analysis of the HFBP-EM from HERA Campaign 6. Careless responses on surveys were identified and flagged. Actigraphy and heart rate data were processed according to best practices, and sleep and heart rate metrics were summarized as daily aggregates. Data from objective performance tasks were transformed, and key performance metrics were extracted. A set of meta-data variables (e.g., mission, participant ID, mission day) were generated to define key variables with which data could be merged across missions for future analyses. Data were merged across multiple missions to inform analyses for the further development of shorter forms of some measures and to add to the HFBP-EM Harmonization project. We delivered a brief report summarizing our findings from HERA Campaign 6 to the HFBP Element (Bell et al., 2023). Overall, all HFBP-EM measures were successfully implemented in Campaign 6, and data yield was high. The most complete data was obtained from behavioral performance measures (99.3%), followed by surveys (98.4%), and physiological markers (94.6%), respectively. The greatest data loss was linked to heart rate data collection in earlier missions. This was largely due to skin irritation reported by some crew members and was significantly mitigated in later missions after introducing alternative electrode patches and a troubleshooting guide to address potential skin irritation. Data from Campaign 6 was processed according to the format established by BHP Lab for our harmonized database intended to integrate all HFBP-EM collected across all analog settings. Second, we developed short-form versions of several measures of behavioral health and team performance, including the Beck Depression Inventory (BDI-II), the Profiles of Mood States (POMS), Group Living Survey, and Team Cohesion, Team Performance (rated by crew), Team Performance (rated by Mission Control Center/MCC), Team Processes, and Team Climate surveys. We followed best practices for survey development by utilizing HFBP-EM data from previous HERA Campaigns 4 and 5 to a) examine content coverage of the existing long form HFBP-EM surveys, b) develop short-forms of measures using a psychometric approach appropriate to each measure and input from subject matter experts (SME), and c) implement short and long form versions in HERA C6. We made progress on the short-form measures validation analyses with HERA C6 data. The updated team performance measure showed improved sensitivity and variability over the previously used team performance measures and was incorporated into the standard HFBP-EM suite. Results for the Group Living Survey showed that the measure had good psychometric properties. A manuscript detailing these results is under review at a peer-reviewed journal. Analyses for additional short-form measures in the HFBP-EM suite are in progress. Third, we made significant progress in preparing for HERA Campaign 7 (C7). We received approval from the Human Research Program for our proposed efforts and changes to the protocol. For HERA Campaign 7, we proposed research for 5 primary objectives: (1) Continue HFBP-EM data collection through HERA Campaign 7, (2) Summarize the extent to which contributing factors (e.g., stressors such as off-nominal emergency events) shape key BMed and Team risk metrics, (3) Determine thresholds for key BMed and Team risk metrics, (4) Characterize changes in social connectedness between crewmembers and their family/friends in relation to communications delay, and (5) Continue data collection for short-form items for measures of Behavioral Health and Teams. Preparation for Campaign 7 Mission 1, which is scheduled to begin in January 2024, is underway.
Bibliography Type:	Description: (Last Updated: 08/01/2025)
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Papers from Meeting Proceedings	Bell ST. "Behavioral Health and Performance for Future Space Exploration." 35th International Space University Space Studies Program, São José Dos Campos, Brazil, June 26-August 25, 2023. 35th International Space University Space Studies Program, São José Dos Campos, Brazil, June 26-August 25, 2023. , Aug-2023